

```

# =====
# Download BLS OEWS Research Estimates – Sector Files
# 2014 and 2015 (to match existing Data 2/2013_sectors/)
# =====
# Files live at: https://www.bls.gov/oes/special-requests/
# Naming pattern: oes_research_{year}_sec_{sectors}.xlsx
# =====

# Spoof a browser user-agent – BLS blocks default R/curl agents
options(HTTPUserAgent = paste0(
  "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) ",
  "AppleWebKit/537.36 (KHTML, like Gecko) ",
  "Chrome/120.0.0.0 Safari/537.36"
))

base_url <- "https://www.bls.gov/oes/special-requests"

# Sector suffixes – same for both 2014 and 2015
sector_suffixes <- c(
  "11-81",
  "21-22-23",
  "31-32-33",
  "42-44-45",
  "48-49",
  "51-54",
  "52-53",
  "55-56",
  "61",
  "62",
  "71-72",
  "99"
)

download_year <- function(year) {
  out_dir <- paste0("Data 2/", year, "_sectors")
  dir.create(out_dir, showWarnings = FALSE, recursive = TRUE)
  message(sprintf("\n=== Downloading %d sector files to %s ===", year, out_dir))

  for (suffix in sector_suffixes) {
    fname <- sprintf("oes_research_%d_sec_%s.xlsx", year, suffix)
    url <- paste0(base_url, "/", fname)
    out_path <- paste0(out_dir, "/", fname)

    if (file.exists(out_path)) {
      message(sprintf(" [skip] %s already exists", fname))
      next
    }

    message(sprintf(" Downloading %s ...", fname))
    tryCatch({
      download.file(url, out_path, mode = "wb", quiet = TRUE)
      message(sprintf(" [ok] %s (%.1f KB)",
        fname, file.size(out_path) / 1024))
    }, error = function(e) {
      message(sprintf(" [FAIL] %s – %s", fname, conditionMessage(e)))
      if (file.exists(out_path)) file.remove(out_path)
    })

    Sys.sleep(0.5) # be polite to BLS servers
  }
}

download_year(2014)
download_year(2015)

```

```
message("\nDone. Verify file counts:")
for (yr in c(2014, 2015)) {
  d <- file.path("Data 2", paste0(yr, "_sectors"))
  n <- length(list.files(d, pattern = "\\\\.xlsx$"))
  message(sprintf("  %d_sectors: %d / %d files", yr, n, length(sectors_suffixes)))
}
```